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Development of a multimodal magnetic resonance imaging-based machine learning prediction model for flight cadets

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In the realm of civil aviation, the existing methods for selecting and training flight cadets have limitations, such as long evaluation cycles and susceptibility to subjective factors. This study integrated multimodal magnetic resonance imaging (MRI) data, including structural MRI (sMRI), diffusion tensor imaging (DTI), and functional MRI (fMRI), with machine learning techniques. The aim was to construct prediction models capable of accurately differentiating flight cadets from ground cadets. Data were collected from 39 flight cadets with extensive flight training and 37 ground cadets. Representative features were meticulously extracted from each modality and fused at the feature level. Four machine learning classification algorithms, namely logistic regression (LR), random forest support vector machine and Gaussian naive Bayes were employed for model construction. Rigorous five-fold cross-validation and permutation tests were conducted to ensure model reliability. The results revealed that the multimodal fusion model (sMRI + DTI + fMRI + LR) exhibited the optimal performance, achieving an accuracy of 0.838, an area under the receiver operating characteristic curve (AUC) of 0.942, a sensitivity of 0.835, and a specificity of 0.834. Through SHapley Additive exPlanations analysis, features with high contributions to the classification were identified, which were closely associated with advanced cognitive functions, visual processing, and attention allocation. This research not only offers a novel approach for the selection and training evaluation of flight cadets but also demonstrates the potential of multimodal MRI-based machine learning models in exploring the neural mechanisms underlying flight-related skills.

Keywords Magnetic resonance imaging, Machine learning, Flight cadets, Multimodality, Civil aviation

With the rapid development of civil aviation, the number of flight missions for civil aviation pilots has increased. The rapid growth of the civil aviation industry has also posed new challenges for pilot training. The selection and training of civil aviation pilots is a complex and time-consuming process, requiring flight cadets to undergo a series of systematic flight training to meet established standards. Currently, the screening and training of flight cadets mainly rely on a limited set of reference indicators, focusing primarily on evaluating physical fitness, psychological endurance, and flight performance, among other aspects, to conduct a comprehensive assessment. However, the existing evaluation methods still have certain limitations. First, the evaluation cycle is relatively long, requiring a significant amount of time to produce results. Second, the evaluation outcomes are highly influenced by subjective factors, which may lead to biased results. Finally, the current methods are inadequate in accurately predicting early on whether a trainee is suitable for long-term flight training. As a result, traditional selection and training methods have not met the aviation industry's demand for efficient and precise pilot selection and development. Therefore, there is an urgent need to introduce more objective, scientific, and efficient assessment indicators and construct more scientifically grounded evaluation models. By employing advanced data analysis techniques to develop machine learning models, comprehensive assessments of cadets can be completed in a shorter period, offering more accurate predictive results. This approach not only optimizes the pilot selection process but also helps flight training institutions better design personalized training programs, thus improving the quality of pilot training and meeting the demands of the rapidly evolving aviation industry.

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Flying is fundamentally a high-level cognitive activity[1], requiring advanced cognitive abilities and rapid responses. Numerous prior studies have indicated that cognitive capabilities are intricately associated with diverse areas of the brain[2–5]. Magnetic Resonance Imaging (MRI), as a radiation-free imaging technique with high soft tissue resolution, has become a valuable tool for exploring the structure and function of the human brain. Structural Magnetic Resonance Imaging (sMRI) provides detailed information about the brain's anatomical structure. In addition to displaying anatomical structures, MRI can also assess brain functional activity, the integrity and connectivity of neural fiber bundles, and the diffusion of water molecules within tissues through techniques such as Functional MRI (fMRI) and Diffusion Tensor Imaging (DTI), providing crucial insights for research on brain function and the mechanisms of the nervous system. Previous studies, based on unimodal brain neuroimaging data from pilots, have identified functional and structural changes in pilots' brains compared to those of the general population[6,7]. Chen et al. found that pilots exhibit enhanced connectivity within the Default Mode Network (DMN) compared to the general population. Additionally, there was decreased intra-network functional connectivity and increased temporal characteristics of functional dynamics[1,8]. Xu et al.[9] analyzed the metric of the amplitude of low-frequency fluctuations (ALFF) extracted from the fMRI data and found that the activity of the brain regions related to visual information processing in pilots was enhanced. Chen et al. conducted a DTI study on flight cadets, which revealed that the flight cadet group exhibited significantly higher fractional anisotropy (FA) values in the left cingulum bundle-cingulate gyrus part (Left CGC) and the right superior longitudinal fasciculus (Right SLF) compared to the control group without flight training[10]. However, previous studies have used only single-modal neuroimaging data to analyze the pilot population, which has limitations in acquiring comprehensive brain information. Combining neuroimaging data of multiple modalities can provide a more comprehensive understanding of the brain. Furthermore, in the field of medical imaging analysis of MRI data, the general linear model (GLM), analysis of variance (ANOVA), and region of interest (ROI) analysis are currently widely used traditional methods for analyzing brain data; however, all three have certain limitations in practical application. In brain imaging research, ANOVA is mostly used to compare differences in brain structural or functional metrics across multiple groups—for instance, analyzing differences in gray matter volume, ALFF values, and other metrics among flight cadets at different training stages. Through repeated-measures ANOVA, it is also possible to track dynamic changes in brain metrics throughout the training process. Nevertheless, this method imposes strict requirements on the normality of data and homogeneity of variance. When dealing with high-dimensional imaging data, it struggles to effectively address issues of false positives or false negatives caused by multiple comparisons. ROI analysis, by contrast, requires defining regions of interest based on prior theories (e.g., brain regions related to flight, such as the visual cortex and prefrontal cortex) or anatomical templates (e.g., the AAL template, Desikan-Killiany template). Subsequent statistical testing is then performed using averaged metrics extracted from these regions—such as the mean FA value and functional connectivity (FC) strength within the ROI. While this method offers strong targeting and intuitive result interpretation, it is highly dependent on the rationality of ROI definition: if pre-defined regions fail to cover potential key brain areas, important neuroimaging difference signals may be directly missed. GLM typically calculates linear correlations at the group level. In reality, however, relationships between brain structure and function, as well as between training effects and neural metrics, often exhibit non-linear characteristics—features that linear models cannot accurately fit. Additionally, GLM rarely accounts for interactions between multiple voxels or variables; yet the implementation of brain function inherently relies on the synergistic interaction of multiple regions and metrics. As medical imaging research advances toward higher dimensionality and greater granularity, such methods based on traditional statistical frameworks have gradually become insufficient to meet researchers' needs for deciphering complex brain mechanisms.

In recent years, there has been a significant increase in the studies that apply machine learning (ML) algorithms to neuroimaging data. This phenomenon is mainly attributed to the excellent modeling flexibility and data processing capabilities of ML algorithms. Compared with traditional analysis methods, ML techniques can efficiently extract features from raw imaging data, have excellent processing capabilities for small sample data, and are more sensitive to subtle changes in neuroimaging data[11,12]. Based on this characteristic, in many brain function studies, researchers have used ML algorithm models to conduct comparative analyses of neuroimaging data from different populations, with the aim of gaining insights into the neural mechanisms of the brains of different populations and further strengthening the exploration of the intrinsic functions of the brain[13]. At present, ML algorithms have been widely integrated into the medical field, and their practicality and adaptability in clinical practice are also showing an increasingly upward trend[14].

Currently, the data collection objects in many brain function studies mostly come from patient populations, such as those with Alzheimer's disease and mental illnesses[15,16]. Since the physiological structure and morphology of the brains of patients have significantly changed compared with those of healthy people, the ML method can efficiently and conveniently capture the brain lesion areas, thus being applied to clinical diagnosis[17]. However, there is still a severe lack of brain function studies on healthy populations such as pilots using the ML method to integrate multiple neuroimaging modalities. In previous studies, researchers constructed prediction models by combining multimodal neuroimaging data and physiological signals and conducted a comparative analysis with models constructed based on single-modal data only. The results showed that the models integrating multimodal information significantly improved in terms of prediction accuracy[18,19]. Based on previous research, we speculate that an ML model using neuroimaging data that integrates three modalities of fMRI, DTI, and sMRI as feature inputs can more accurately distinguish between the group of flight cadets and the ground cadets without flight training compared with models with single-modal feature inputs.

For neuroimaging data of different modalities, according to their imaging characteristics, we extracted their respective representative indicators as feature inputs for the ML model. In this study, the functional connectivity (FC) matrix was extracted based on rs-fMRI, and the mean diffusivity (MD), fractional anisotropy (FA), axial diffusivity (AD), and radial diffusivity (RD) were extracted based on DTI. Since sMRI reflects various brain

morphological features, such as gray matter volume, gray matter density, cortical thickness, and cortical surface area, which are difficult to observe and analyze with the naked eye, based on this characteristic, we extracted radiomics and deep learning features for sMRI. Prior research has demonstrated that by means of radiomics and deep learning, numerous quantitative features can be retrieved from morphological structure images, including shape, texture, signal intensity distribution, as well as the identification and learning of complex patterns and implicit information in the images[20–22].

Therefore, the purpose of this study is to develop an ML model based on neuroimaging data to accurately distinguish between flight cadets and healthy ground cadets without flight training. We used four ML classification algorithms for modeling, namely logistic regression (LR), random forest (RF), support vector machine (SVM), and Gaussian naive Bayes (GNB). We carried out horizontal comparative analyses to assess how single—modal and multi-modal fusion metrics perform in terms of crucial performance indicators, namely accuracy, sensitivity, and specificity, when applied to different classifiers. In addition, by analyzing the SHapley Additive exPlanations (SHAP) values of the best model and the features that have a high weight contribution to the classification decision, we explored new evaluation indicators for flight training from the perspective of brain neuroimaging.

Materials and methods

Participants

The participants were selected from the Civil Aviation Flight University of China(CAFUC). The recruitment announcement detailed the experiment's content, screening criteria, and potential hazards and risks. A total of 79 cadets participated in the MRI scanning: 40 flight cadets, all of whom had completed at least 230 h of flight training at the CAFUC, obtained private and commercial pilot licenses issued by the Civil Aviation Administration of China, and were qualified to independently fly small airplanes. The remaining 39 subjects formed a control group consisting of cadets from the ground program. In this study, both the experimental group and the control group were composed of cadets from the same enrollment batch. These cadets had similar entrance examination scores. Throughout their undergraduate studies, the learning content, total credit hours, and theoretical class hours of the two groups of cadets were essentially equivalent. In the initial selection of flight cadets, the emphasis was on physical fitness and visual acuity, and the selection requirements in other aspects were the same as those for ground cadets. The control group consisted of cadets who received no flight training and only participated in ground-based theoretical courses (such as aviation theory, meteorology, and navigation principles). All ground-based cadets came from the engineering discipline of the same institution, and possessed neither a flight license nor any experience in flight simulation. Their training content significantly differed from that of the flight cadets, focusing primarily on theoretical learning and ground operation skills, rather than practical flight tasks. The recruitment period for participants commenced in June 2022 and concluded in November 2022. The acquisition of magnetic resonance imaging data began in June 2022 and ended in November 2022.

Inclusion criteria for all subjects were: (1) no metal objects in the body; (2) in good health at the time of participation; (3) all male; (4) aged 21–25 years; (5) right-handed; (6) no history of psychiatric disorders; (7) no drug or alcohol dependence; (8) no history of traumatic brain injuries, cerebrovascular disease, or chronic pain; (9) all holding undergraduate degrees in engineering; (10) All participants provided informed consent and voluntarily signed a consent form before the experiment commenced. The research adhered to the principles of the Declaration of Helsinki. The Ethics Committee of the University of Electronic Science and Technology of China (Chengdu, China) approved the experimental procedures under No. 1420200408-07.

Data acquisition and preprocessing

All data were collected using a 3-Tesla MRI scanner. The T1-spoiled gradient recalled echo pulse sequence was employed with scan parameters that included a repetition time (TR) of 5.952 ms, an echo time (TE) of 1.964 ms, a flip angle of 9°, a matrix size of 256 × 256, a slice thickness of 4 mm with no gap, a field of view (FOV) of 25.6 cm × 25.6 cm, and a total of 154 slices were acquired, the voxel size = 1 mm × 1 mm × 1 mm. The functional images at rest were obtained using a gradient-echo echo-planar imaging sequence. Subjects were instructed to keep their eyes closed for the 510-s scan while remaining awake. The scan parameters were: TR = 2000 ms, TE = 30 ms, flip angle = 90°, matrix size = 64 × 64, slice thickness = 4 mm with a 0.4 mm gap, FOV = 24 cm × 24 cm, 35 slices per volume, and a total of 255 volumes acquired, the voxel size = 3 mm × 3 mm × 3 mm. DTI acquisition parameters. A single—shot spin—echo planner sequence was used to acquire diffusion tensor images with 64 gradient directions. The number of excitations (NEX) was 2, the matrix size was 128 × 128, the field of view (FOV) was 25.6 cm × 25.6 cm, the slice thickness was 2 mm, the b-value was1000 s/mm^2 ,the repetition time was 8500 ms, and 75 axial slices of the whole brain were acquired without spacing.

The T1 structural images underwent preprocessing with the help of the Statistical Parametric Mapping Toolbox (SPM12) and Computational Anatomy Toolbox (CAT12) on the basis of MATLAB 2022b. The following steps were involved: (1) Quality assessment and orientation rectification of the original images; (2) Segmentation of the images into gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF); (3) Registration and modulation among different subjects; (4) Spatial normalization to the Montreal Neurological Institute (MNI) space with a voxel size of 1.5 mm × 1.5 mm × 1.5 mm; (5) Application of spatial smoothing using an 8–mm full-width at half-maximum Gaussian kernel; (6) Estimation of the total intracranial volume for the purpose of correcting individual variances in brain size and volume.

The RestPlus toolbox on the MATLAB 2022b platform was utilized for preprocessing resting-state fMRI data. The pre-processing steps for fMRI data include: (1) removal of the first five time points to minimize interference from unstable magnetization vectors; (2) temporal slice correction to mitigate scan time differences between slices; (3) head motion correction by excluding samples with translation > 2 mm or rotation > 2°; (4) regression of covariates, including average CSF signal, average WM signal, and head motion signals derived from the

Friston-24 parameter model; (5) Realigning individual brain images to the standard space of the MNI at a resolution of 3 mm × 3 mm × 3 mm through spatial normalization; (6) Spatial smoothing processing was carried out using a Gaussian filter to reduce spatial noise, and signals outside the frequency range of 0.01–0.08 Hz were removed; (7) removal of linear drift.

The DTI data underwent preprocessing with the utilization of the FSL software[23], which included: (1) converting the image format; (2) removing non-brain structures; (3) correcting head motion and eddy currents; (4) applying the DTIFIT program to calculate the DTI parameters.

Extracting MRI metrics

The FC metrics of the fMRI data were computed by means of the RestPlus 1.28 toolkit, SPM12, and self–written code on the MATLAB 2022b platform. In particular, based on the automated anatomical labeling (AAL) atlas[24], the FC matrix of 116 brain regions (as presented in Table 1, which consists of 90 cerebral and 26 cerebellar regions) was extracted. The blood oxygen level—dependent (BOLD) signals of 116 brain regions were extracted from the pre-processed resting-state fMRI image data. These signals consisted of 250 time points, thus forming a BOLD time-series signal for each subject. FC is defined as the correlation between the average blood oxygen level-dependent (BOLD) time series of brain regions, and this correlation is measured using the Pearson correlation coefficient. To meet the requirements of subsequent analyses and statistical tests, the Fisher's r-to-z transformation needs to be applied to convert the obtained correlation coefficients into z-values. After these steps, a symmetric 116 × 116 FC matrix was obtained for each subject in this study. Then, the duplicate elements of the symmetric matrix were removed, and ultimately a total of (116 × 115)/2 = 6,670 FC features were obtained. Figure 1 shows the FC matrix of one subject. FC reflects the synchronization of signal activities between two brain regions and is often used as a feature in machine learning research[25].

Currently, for DTI data, methods based on voxel-based morphometry (VBM)[26] or tract—based spatial statistics (TBSS)[27,28] are commonly used for analysis. The analysis results obtained by these methods can reflect the changes in the white matter microstructure and the integrity of the fiber tracts in the subjects' brains. However, the former has problems with registration errors and smoothing, while the latter can only evaluate the average diffusion changes of the entire fiber tract and lacks precise positioning for individuals. The Automated Fiber Quantification (AFQ) method effectively overcomes the limitations of traditional methods in fiber tract imaging and diffusion parameter estimation[29,30]. Relying on the whole—brain deterministic fiber tract imaging technique, this method systematically reconstructs the fiber tracts. On this basis, AFQ selects 100 anatomically equivalent positions on the fiber trajectories and estimates the diffusion parameters point-by-point for each specific fiber bundle[31]. We used the AFQ technique to extract the relevant metrics of the Diffusion Tensor DTI data. The specific steps are as follows: First, Using the AFQ technique, the fiber tracts in the brain were traced and reconstructed. Then, the reconstructed fiber tracts were registered with the Montreal Neurological Institute (MNI) template, and a total of 20 fiber tracts were obtained. Then, the white matter fiber tracts were tracked, with the tracking threshold set as FA < 0.2 or the fiber trajectory angle > 30°. One hundred equidistant segments were determined on each fiber tract, and the core position of the fiber tract was determined according to the average position of each segment. The weighted average on the fiber tract was utilized to calculate the diffusion tensor eigenvalues of each segment.

Based on the sMRI data, we used the "PyRadiomics" software package to obtain radiomics (Rad) features. In the task of feature extraction using deep learning, we built a deep convolutional neural network based on 3D ResNet. Based on the 3D ResNet-18 model, we removed the fully connected layer that finally outputs the classification result in the convolutional neural network[32]. In this way, we can obtain the intermediate features output by the second-to-last layer of the model instead of the classification result. The construction of the overall dataset in this study is shown in Fig. 2.

Feature selection

This paper combines the two-sample t-test method and the least absolute shrinkage and selection operator (LASSO) to screen sample features. The two-sample t-test is an algorithm frequently used in MRI, and it has also been applied to feature screening in some neuroimaging studies using ML algorithms[33], achieving good results. In the context of feature selection, The two-sample t-test can identify features with significant differences from two sets of independent sample data. However, a considerable number of the features screened by this method have limited contributions to the prediction results. To solve this problem, make the feature screening more accurate and reduce redundancy, this study combines the LASSO algorithm to carry out further feature screening work. By means of its unique regularization mechanism, LASSO drives the coefficients of some unimportant features to zero, thus achieving the goal of feature sparsity. Finally, this study combines the two

Characteristics	Flight cadets (N = 39)	Controls (N = 37)	*T*-value	*P*-value
Age (years)	22.9 ± 0.96	22.4 ± 0.95	2.162	0.034
Gender (% male)	100	100	–	–
Education (years)	16	16	–	–
Handedness(% right)	100	100	–	–
Total flight time (hours)	240.69 ± 8.62	0	–	–

Table 1. Comparison of demographic data between the two groups.

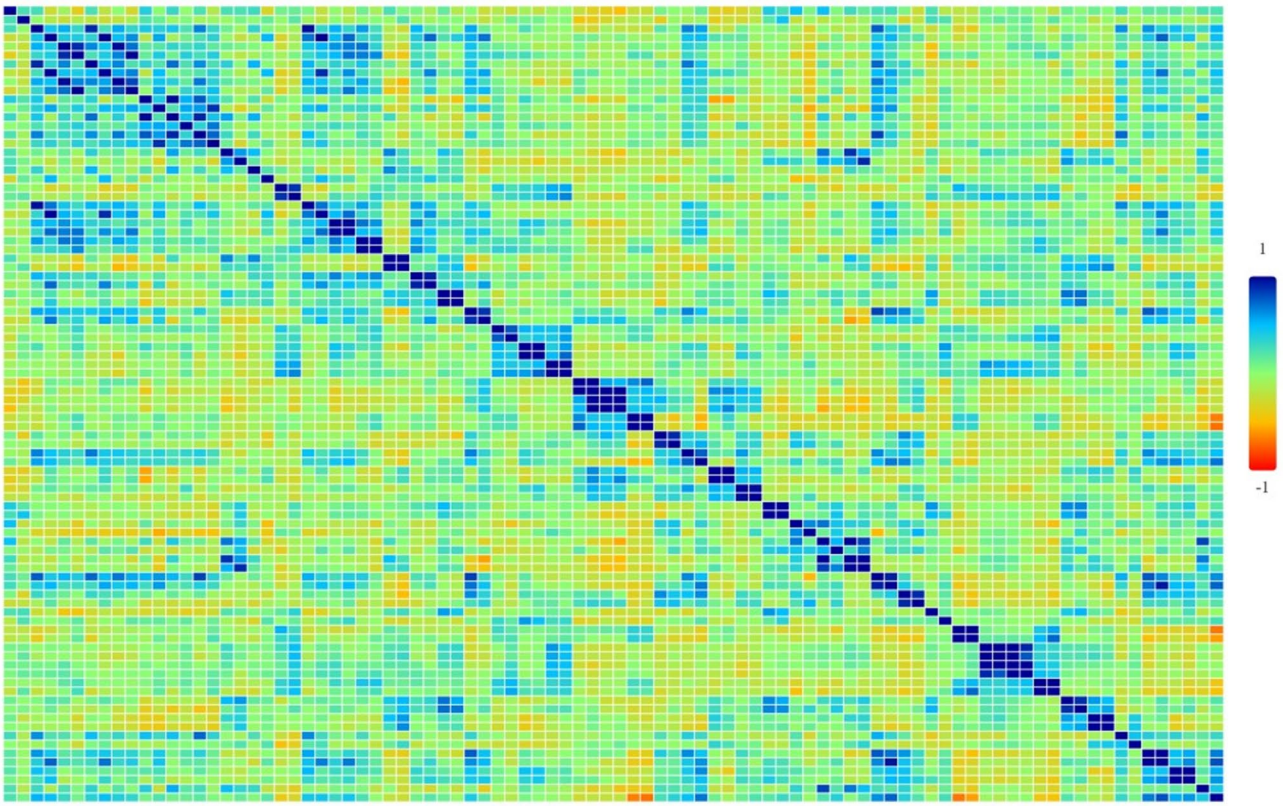


Fig. 1. Functional connectivity matrix.

feature screening algorithms and successfully constructs a feature dataset that can be directly used for the training of machine learning models. In this study, code was written using Scipy (version 1.11.1) package to implement the mentioned feature screening algorithm.

ML model establishment

In this research, four ML classification algorithms are utilized to develop models. The ML algorithms applied in the construction of classification models are Logistic LR, RF, SVM, and GNB. All prediction models were created using Python 3.9, and these models rely on the "sklearn 0.22.1" package. Grid search was carried out to find the optimal hyperparameters for each of these four algorithms. The dataset was randomly divided into training and testing sets at an 4:1 ratio.

Model evaluation and validation

This research utilized frequently—used evaluation metrics including the Receiver Operating Characteristic (ROC) curve, the Area Under the ROC Curve (AUC), accuracy, sensitivity, and specificity to evaluate the predictive capabilities of the models. To guarantee the stability of the models, five-fold cross-validation was used to assess the generalization capacity of the models. On this basis, the process was iterated 100 times to obtain the average ROC curve. Subsequently, ROC curve analysis was performed to quantify the model discrimination, and the obtained AUC and confusion matrix were used to evaluate its prediction accuracy. Machine learning has the property of a "black box". To better interpret the results of ML, we applied the SHAP values to explain the predictions made by the most effective model and plotted the average importance of each feature and the SHAP values. In addition, a permutation test was used to evaluate the statistical significance of the model's predictions, with the number of permutations set to 5000 times.

Results

Participant demographics

In the experiment, 3 subjects were removed from the study due to being left—handed or having excessive head movement. Ultimately, MRI data from 39 flight cadets and 37 ground cadets were incorporated into the study. The demographic data of the subjects ultimately included in the study is presented in Table 1.

Feature screening and fusion

In this study, three types of neuroimaging data metrics were extracted. The multimodal data fusion technique employed was feature-level fusion, also known as pre-fusion. Pre-fusion, also known as feature-level fusion, is a commonly used strategy in multimodal data fusion. It refers to a process where, during the data preprocessing

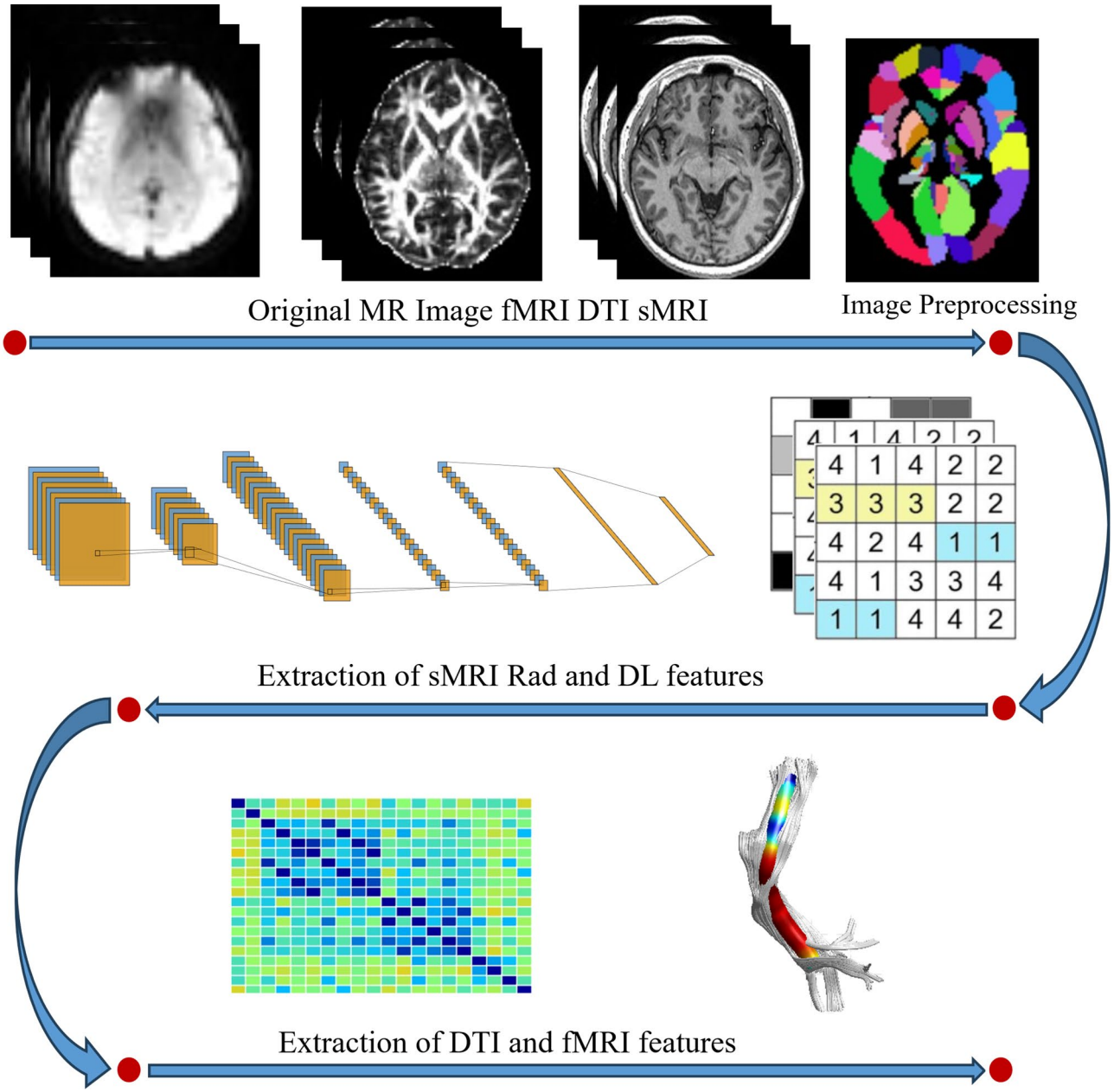


Fig. 2. Construction of the MRI feature dataset.

stage, feature vectors obtained from different modalities (e.g., sMRI, DTI, fMRI) after feature extraction and screening are directly concatenated and integrated to form a unified multimodal feature matrix. This matrix is then input into a ML model for training[32]. By integrating the features of different modalities, the complementary information between the modalities can be fully utilized, which leads to an enhanced understanding of the data by the model[32]. Since the various metrics used were derived from neuroimaging data of different modalities, there were differences in the quantity and scale of the extracted metrics. Therefore, when conducting the two-sample t-test, we applied different threshold levels (p-value) to adjust and control the number of features ultimately fed into the ML model. The p-value is an important indicator for measuring whether a certain feature shows a significant difference between groups. When selecting the p-value threshold, we referred to common significance levels (such as 0.05), which is a widely used standard for distinguishing statistically significant and non-significant features. However, with the advent of machine learning and high-dimensional datasets, an increasing number of studies have recommended using stricter thresholds. For instance, the study by Madsen et al.[34] proposed that when analyzing neuroimaging data, a lower p-value threshold could more effectively reduce false positives caused by the multiple comparison problem. To improve the precision of feature selection and avoid incorporating noisy data as significant features into the model, we gradually adjusted the p-value threshold to stricter levels (such as 0.01 and 0.005). This multi-level selection approach enabled us to gradually control the false positive rate resulting from multiple comparisons while maintaining the interpretability of the data. Table 2

	FC	sMRI	DTI	sMRI + FC + DTI
p	0.01	0.05	0.01	0.005
two-sample t-test	141	61	129	40
two-sample t-test + LASSO	24	14	21	16

Table 2. The threshold levels set for each metric and the number of features after being screened by the two-sample t-test and LASSO.

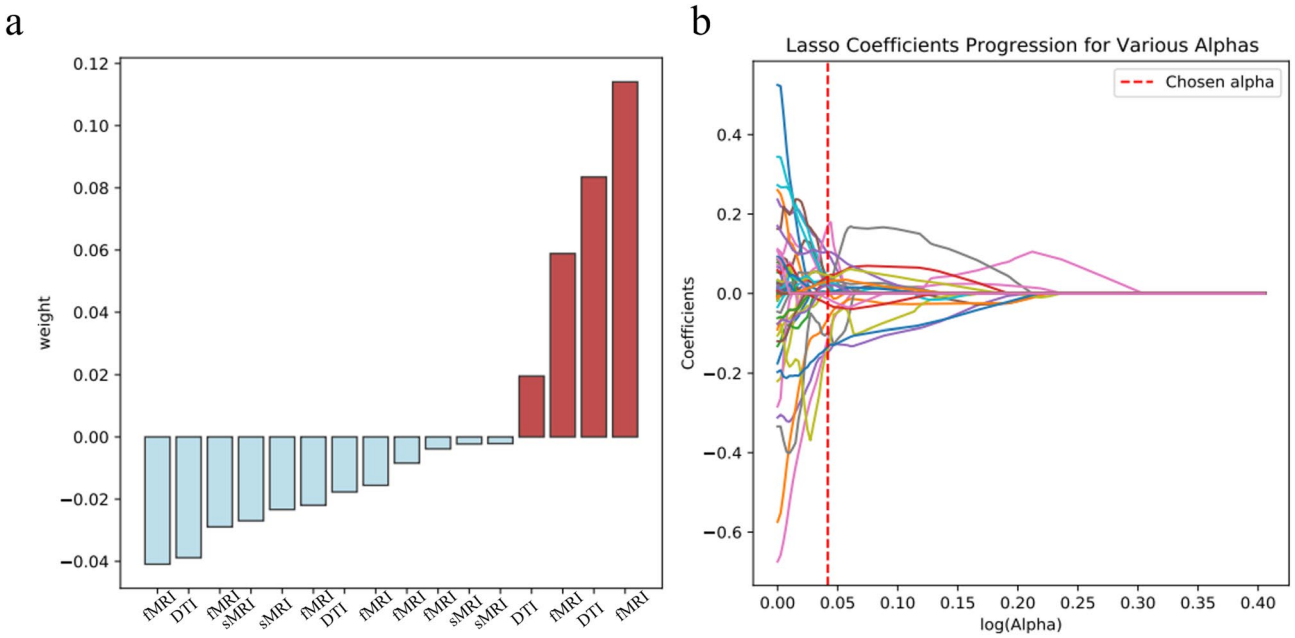


Fig. 3. (**a**) The number, weights and sources of the features determined by LASSO regression for the pre-fused features. (**b**) LASSO coefficient path.

presents the screening threshold levels for each modal metric, along with the number of features sequentially selected by the feature selection algorithm. Figure 3a shows the number of features determined after the pre-fused features were selected through LASSO regression and the modalities to which these features belonged. Figure 3b is a graph of coefficient paths for LASSO regression.

Performance comparison of classifiers for various metrics

In this section, this study evaluated the performance of different modal metrics on multiple classifiers and conducted a horizontal comparison of the performance evaluation metrics of each modal metric in different classifiers. By screening the features of the neuroimaging metrics of three single modalities and the fused modality, an ML prediction model was constructed in this study based on the optimal feature subset. The findings indicated that, under the same modal metrics, the classification performance of the GNB and LR classifiers was generally better than that of the SVM and RF classifiers. After comprehensively comparing the modal metrics, the ML model that used the metrics integrating the three modalities as feature inputs performed the best in terms of classification performance.

As shown in Fig. 4, the AUC of the three modal metrics using pre-fusion were all greater than those of the single modality, and the classification effects of the four ML models were significantly higher than the results of random classification. Among them, the LR classifier had the highest accuracy and AUC (83.80%,0.94). In the ML models with single-modal neuroimaging data as feature inputs, the overall classification effect of the sMRI features on the four ML models was higher than the results of random classification. Among them, the GNB classifier had the highest accuracy and AUC (77.30%, 0.86). The overall classification effect of the fMRI features on the four ML models was higher than the results of random classification. Among them, the GNB classifier had the highest accuracy and AUC(71.50% , 0.83). The overall classification effect of the DTI features on the four ML models was higher than the results of random classification. Among them, the GNB classifier had the highest accuracy and AUC, which were 71.50% and 0.76 respectively. In Table 3, we have listed in detail the classification performance evaluation metrics of four machine learning models when data from different modalities are used as feature inputs. Through comprehensive comparison, it was found that the ML classification performance based on the sMRI + DTI + fMRI metrics performed the best overall. Among them, the GNB classifier could achieve

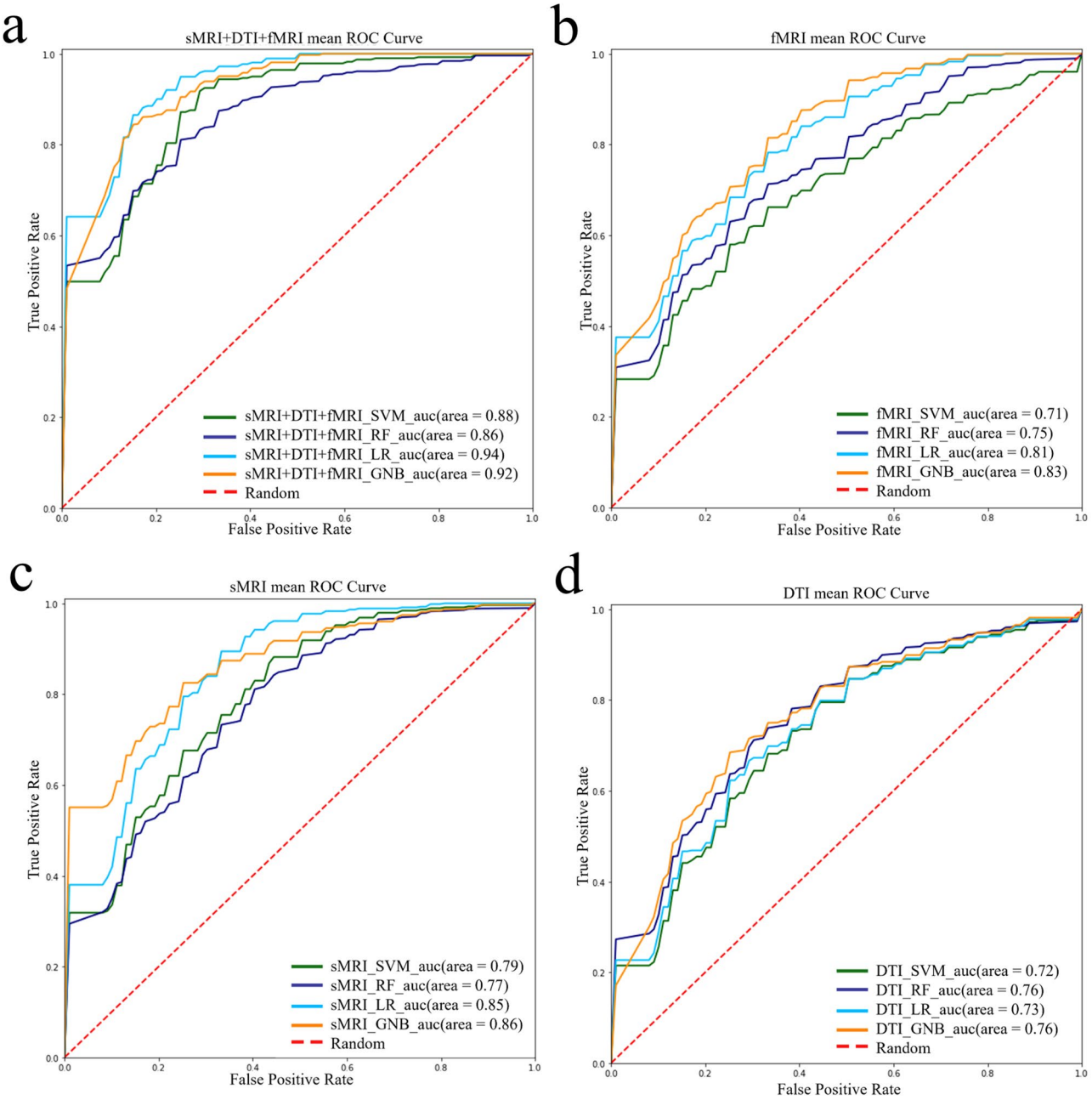


Fig. 4. ROC curves of the metrics of three neuroimaging modalities and the pre-fused metrics.

the highest sensitivity (0.87), and the LR classifier could achieve the highest accuracy, AUC, and specificity, which were 83.8%, 0.94, and 0.83 respectively.

This study adopted the permutation test method to determine whether the model performance was obtained merely by chance. The specific approach is to randomly permute the labels in the dataset to obtain a random dataset. If the performance of the model significantly deteriorates on the random dataset, it indicates that the model has indeed learned effective patterns from the original data. As shown in Fig. 5, we conducted permutation test verification based on the ML models with the optimal classification performance for each modality. The permutation test (number of repetitions: 5000 times) showed that the actual classification accuracies of the optimal ML models with each modality as feature input were all higher than those of the random labels ($p < 0.05$). This indicates that the results of ML modeling are statistically significant and the classification performance is reliable. After comparing various evaluation metrics, it was found that the overall best model was the LR model based on sMRI + DTI + fMRI.

ML model interpretation

Based on the previously obtained best model (sMRI + DTI + fMRI + LR), we further analyzed the best model using SHAP values. The SHAP technique is a powerful tool for interpreting the prediction results of machine

Feature	Model	ACC	AUC	Sensitivity	Specificity
fMRI	LR	0.685	0.803	0.693	0.749
	GNB	0.715	0.826	0.752	0.721
	RF	0.698	0.765	0.747	0.695
	SVM	0.636	0.718	0.678	0.657
sMRI	LR	0.773	0.852	0.843	0.728
	GNB	0.782	0.865	0.837	0.725
	RF	0.692	0.773	0.746	0.665
	SVM	0.712	0.794	0.731	0.706
DTI	LR	0.685	0.735	0.681	0.674
	GNB	0.715	0.764	0.705	0.722
	RF	0.698	0.761	0.718	0.693
	SVM	0.636	0.723	0.686	0.615
fMRI + sMRI + DTI	LR	0.838	0.942	0.835	0.834
	GNB	0.831	0.921	0.850	0.812
	RF	0.773	0.875	0.772	0.780
	SVM	0.803	0.883	0.827	0.792

Table 3. Classification performance of four ML algorithms with different modal data.

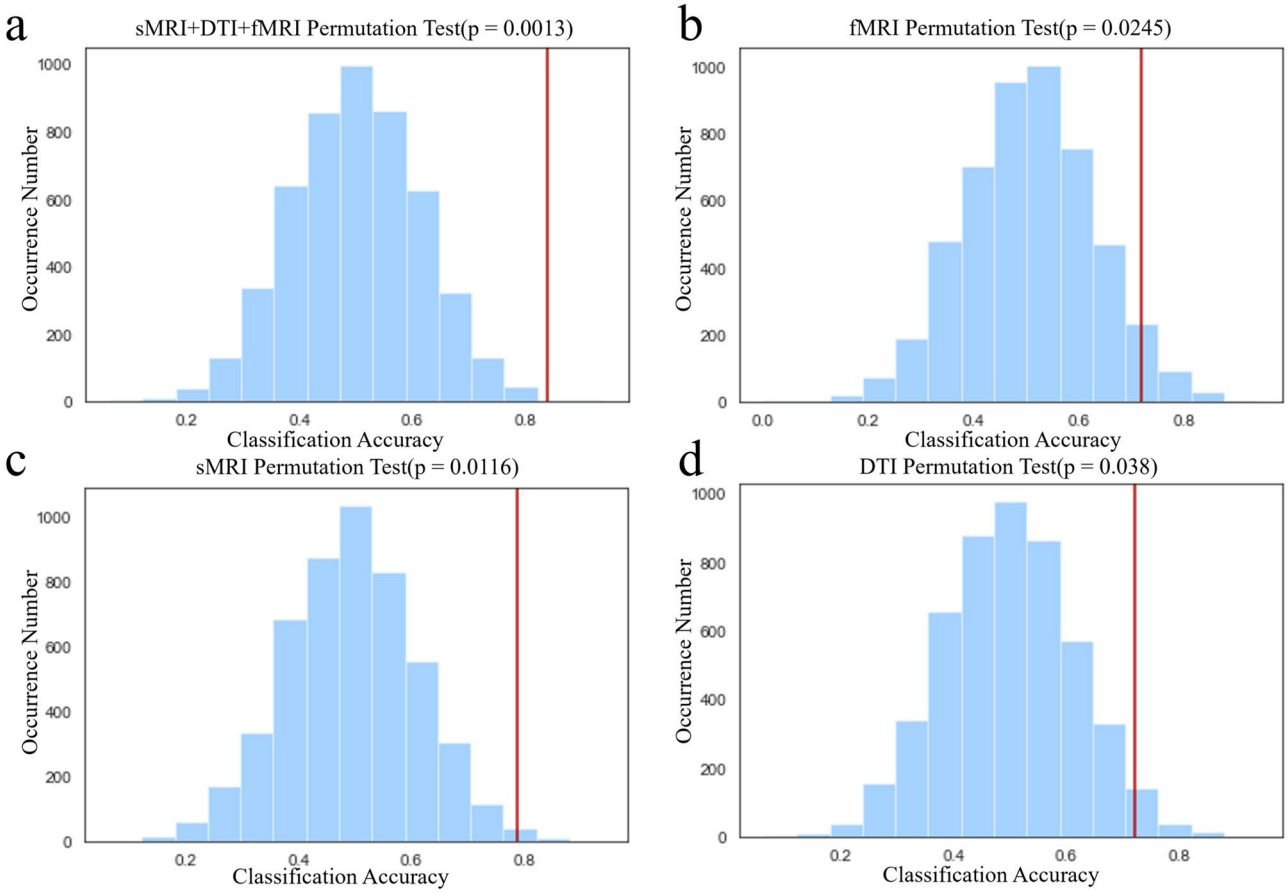


Fig. 5. Results of the permutation test based on the optimal model for each modal metric.

learning models. Based on the concept of Shapley values in cooperative game theory, it is capable of measuring the contribution degree of each feature to the prediction results of the model. The core idea of SHAP values is to regard the model prediction as a cooperative game, where each feature is a participant in the game. SHAP can quantify the importance of each feature and provide consistent and fair explanations[35]. As shown in Fig. 6, we calculated the SHAP values of the 16 features that had been screened in the optimal model (fMRI + DTI + sMRI + LR), and

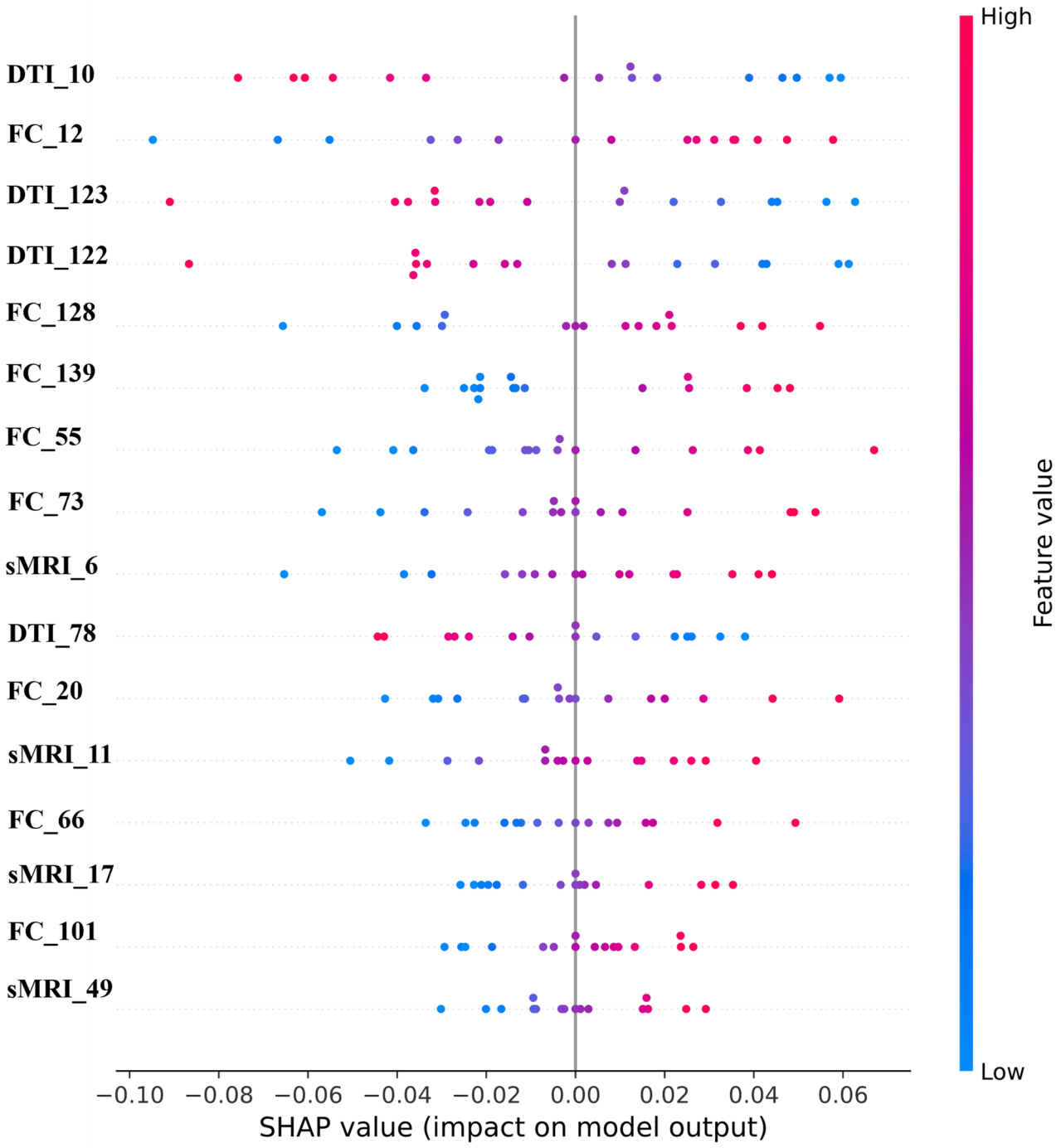


Fig. 6. Scatterplot of SHAP values versus feature values based on the best model, showing the effect of individual features on the model prediction results.

used the SHAP values to intuitively display the impact distribution of each feature on the classification results. In Fig. 6, the contribution of 16 features to the prediction results decreases from top to bottom. Each point in the scatter plot represents a sample, with the horizontal axis indicating the SHAP value of that feature[36]. The positive or negative sign of the SHAP value indicates a positive or negative correlation between the feature and the predicted value of the sample. The positive or negative sign of SHAP values only indicates the direction in which a feature drives the prediction result. A negative value means the feature tends to push the prediction toward the "ground cadet" category, while a positive value pushes it toward the "flight cadet" category. The larger the absolute value of the SHAP value, the more significant the feature's contribution to the prediction. According to the results in Fig. 6, all features extracted from DTI are negatively correlated with the predicted values, while the features from fMRI and sMRI had a positive correlation effect on the predicted values. Figure 7

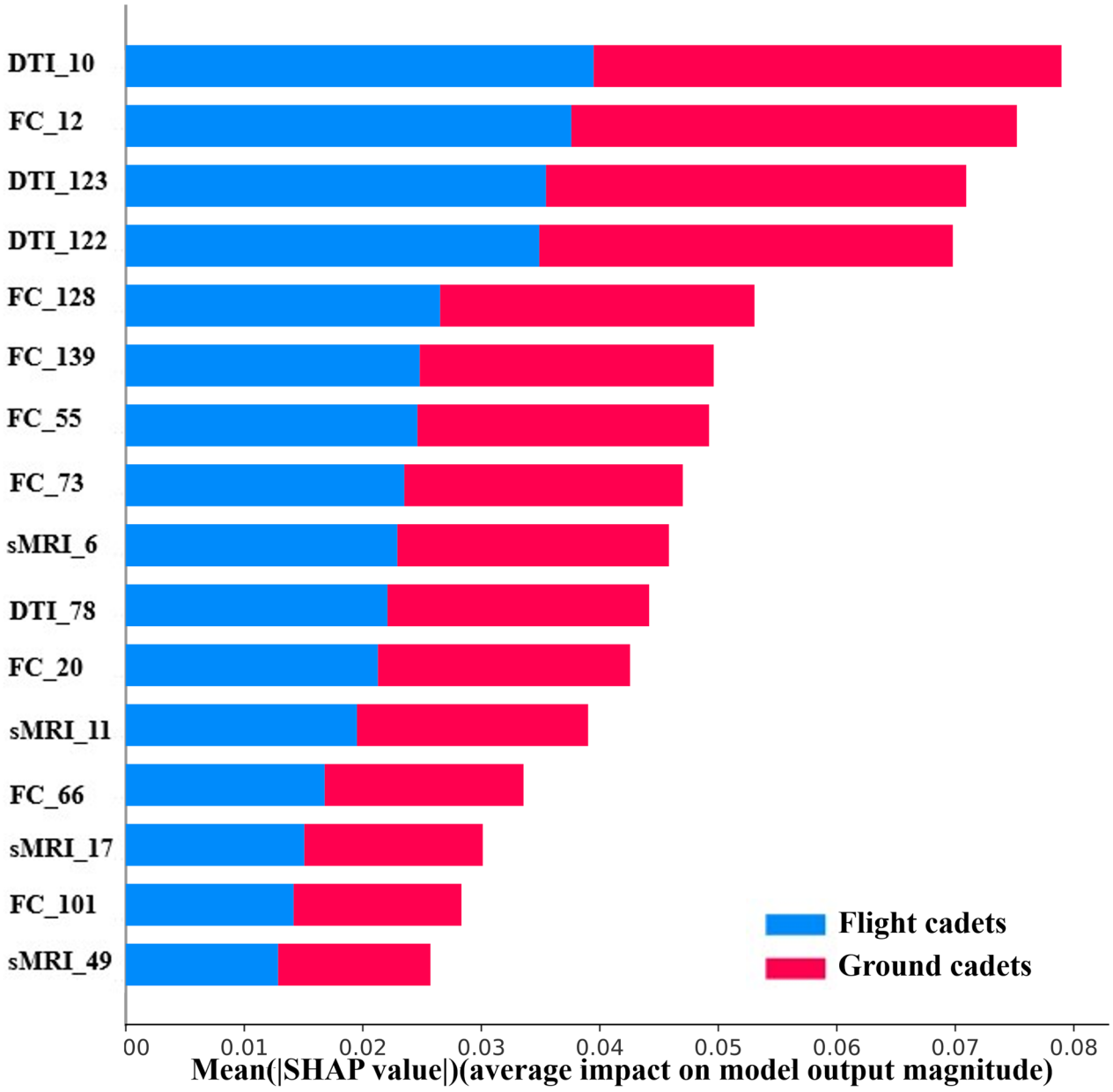


Fig. 7. SHAP value of feature importance based on the best model.

is a summary graph of feature importance, and the importance of features increases from bottom to top. As shown in Fig. 7, all features seem to contribute equally to distinguishing between the two categories of flight cadets (label = 1) and ground cadets (label = 0), because each feature basically occupies 50% of the rectangle. This indicates that the contribution of each feature to the classification of the two categories is relatively average, rather than being contributive only when classifying a single specific category. Since the fused features involve data from three different modalities, the data structure is relatively complex. In order to be able to better trace the modality to which the features with high contribution degrees belong and their specific locations in the brain in the later stage of the study, we recorded the array subscripts of the original data at the initial stage of the study and re-encoded the names of the features after the screening algorithm. The purpose of doing this is that after completing all the modeling processes and interpreting the results, we can trace each feature back to its specific corresponding modality and detailed index according to the saved original array subscripts. This approach helps us better understand the origin of the features and can provide clearer and more accurate interpretations in subsequent neuroimaging data analysis. Table 4 lists the modalities and specific metrics corresponding to the feature names.

Feature name	Modal	Metric
DTI_10	DTI	FA value of the right thalamic radiation tract
FC_12	fMRI	FC value between SFGdor.L and OLF.R
DTI_123	DTI	MD value of the right superior longitudinal fasciculus
DTI_122	DTI	MD value of the left superior longitudinal fasciculus
FC_128	fMRI	FC value between SFGmed.L and PCG.R
FC_139	fMRI	FC value between PUT.R and MTG.L
FC_55	fMRI	FC value between LING.L and PAL.L
FC_73	fMRI	FC value between PAL.R and MTG.L
sMRI_6	sMRI	Radiomics feature: wavelet_2
DTI_15	DTI	FA value of the cingulate gyrus part of the left cingulum bundle
FC_20	fMRI	FC value between HIP.L and PoCG.L
sMRI_11	sMRI	Radiomics feature: wavelet_7
FC_66	fMRI	FC value between MFG.L and OLF.R
sMRI_17	sMRI	Radiomics feature
FC_101	fMRI	FC value between AMYG.L and PoCG.L
sMRI_49	sMRI	Deep learning feature

Table 4. The modalities to which the features belong and the specific corresponding metrics.

Discussion

We extracted representative physiological metrics from neuroimaging data of three different modalities, and adopted the pre-fusion technique to fuse the data of different modalities. Based on the fused multimodal features and single-modal features, we constructed four machine learning (ML) classification models respectively. Through comparative analysis of the performance of each model, we developed and validated a variety of prediction models capable of distinguishing between flight cadets and ground cadets. Among them, the best combination form of the prediction model is fMRI + sMRI + DTI + LR.

The subjects of this study are flight cadets. The flight training that flight cadets undergo is not only training in physical fitness and driving proficiency, but more importantly, it is training in cognitive abilities in a variety of complex environments, such as decision-making, hypothesis, and reasoning during the flight process. Previous studies have proven that cognitive training can affect brain functional factors and change brain morphological characteristics to a certain extent[37,38]. Previous studies have shown that individuals who have received special skill training exhibit changes in brain structure and the functions of brain regions compared to ordinary people. Bermudez et al. found that musicians have a significant increase in the GM of the right lateral surface of the superior temporal gyrus compared with ordinary people[39]. Huang et al. found that compared with the general control group, the GM density of world champion gymnasts increased significantly[40]. As a type of special skill training, the repeated and sudden actions during the flight training process will continuously stimulate the brain regions related to emotional control, cognitive function, operational response, etc. Causse et al. found that during flight missions, when pilots make decisions, the activity of certain brain regions becomes more active, which positively affects their decision—making ability[41]. Specifically, compared to cadets who didn't undergo flight training, those who participated demonstrated significantly elevated activity levels in brain regions associated with emotion regulation and cognitive function. Chen et al. found that the FA values of the right superior longitudinal fasciculus (Right_SLF) and the cingulate gyrus part of the left cingulum bundle (Left_CGC) in pilots are higher than those in people without flight training[10]. Thus, it can be seen that after experiencing systematic flight training, the structure and function of pilots' brains have undergone significant changes. These changes can be analyzed and evaluated through imaging metrics such as fMRI, DTI, and sMRI[9,10,42].

In this study, we not only extracted traditional neuroimaging metrics, such as FC, MD, FA, etc., but also combined radiomics and deep learning features. Radiomics features include morphological features, gray-level statistical features, wavelet features, and so on, while deep learning features are automatically learned from image data through deep learning models. As the network layers deepen, deep learning can progressively extract more advanced and abstract features, such as the local structure and overall shape of objects[43]. By analyzing the SHAP values of the features (as shown in Fig. 6), we found that there is a correlation between the influences of brain functional and structural features on the prediction results. Combining with the physiological meanings contained in the neuroimaging metrics, these findings may indicate that compared with ordinary ground cadets, the brains of cadets who have received flight training have experienced changes in certain functions and morphological structures. These changes may be closely related to the advanced cognitive functions required for flight missions, such as decision-making, spatial navigation, and rapid response abilities. Furthermore, the wide distribution of SHAP values reflects that the contribution of this feature to the prediction result varies across different individuals, which may stem from inter-individual differences in neural plasticity or heterogeneity in training experiences. Despite these individual variations, the model still maintains stable classification performance through fivefold cross-validation and permutation tests ($p < 0.05$). This indicates that the overall trends of these features—such as the negative contribution of DTI features and the positive contribution of fMRI features to flight cadet classification—are reliable, and individual differences have not affected the core predictive ability of the model. Regarding the metrics from fMRI, we adopted the method of brain network partitioning to

further explore the functional organization of the brain in depth. As pointed out in previous studies, the various parts of the brain are not isolated but form a complex network structure through large-scale interconnections. This network approach to the brain helps us understand how brain functions are organized[44] . In previous studies, various brain network partitioning templates have been proposed based on different research perspectives and objectives. To more accurately analyze the brain regions associated with FC features in classification tasks, this study adopts the network template proposed by THOMAS YEO et al.[45] and combines it with the brain region network partitioning method (K-Means clustering) developed by SUN et al.[17]. The AAL brain regions corresponding to the FC features that significantly contribute to the classification results are classified into seven brain networks. As shown in Fig. 8, among the FC features that contribute highly to the prediction results, the features from the frontoparietal network (FPN) and the DMN account for the largest proportion overall. The features from the visual network (VN) and the somatomotor network (SMN) also account for a relatively large proportion, and there are no FC features from the Cerebelum (Cer), the dorsal attention network (DAN), and the limbic network (LN). In the field of brain function research, the DMN has been proven to be closely related to cognitive processes such as self-referential thinking, memory recall, and the theory of mind. The FPN is related to executive functions, working memory, and cognitive control, the VN takes charge of processing visual information., and the SMN is associated with body sensation and motor control. The significant contribution of the FC between these brain networks to the model may stem from the requirement of flight training to quickly integrate and screen multi-dimensional information in a complex and dynamic environment. This process not only requires flight cadets to maintain a high level of attention and effective allocation of attention resources continuously, but also requires them to have a keen perception and encoding ability for external information. In this process, the DMN, VN, SMN, and FPN of flight cadets may be repeatedly activated, thus affecting the activation levels of each brain network and the FC between the networks.

It can be found from Fig. 6 that there is a negative correlation between the features from the DTI modality and the predicted categories. This may indicate that the fiber tracts of cadets after flight training have changed compared to those of ground cadets. Figure 9 shows the fiber tracts where the DTI features with high contributions are located and their FA values. The FA value represents the degree of anisotropy of water molecule diffusion in the brain. A higher FA value usually indicates that the fiber tract has a stronger directionality and a more complete structure[46,47]. While MD reflects the average ability of water molecules to diffuse freely in three-dimensional space, and to a certain extent, it can reflect the microscopic structure of the tissue and the integrity of the myelin sheath[48,49]. We conducted a traceability analysis of the highly contributive features from the DTI modality and found that certain DTI physiological indices of the following fiber tracts had changed compared with the control group: the FA value of the right thalamic radiation (TR), the MD value of the right superior longitudinal fasciculus (SLF) and the left superior longitudinal fasciculus (SLF), as well as the FA value of the left cingulum bundle (CGC). These changes may indicate differences in the structure of these fiber tracts or the integrity of myelin sheaths between flight cadets and ground cadets. Previous studies have shown that in tasks that require attention allocation, the TR can affect an individual's reaction speed. The attention control ability will be enhanced as the FA value at the TR increases, showing a positive correlation[50] , and this FA value is also closely related to the visual search ability[51]. According to the SHAP values, the white matter fiber tract of the CGC is located in the cingulate gyrus of the brain. Its FA value is related to overall memory and cognitive function. Previous studies have revealed that when its FA value level decreases, it will lead to a decline in relevant test scores[52]. The SLF originates from the prefrontal region and connects the frontal lobe, parietal

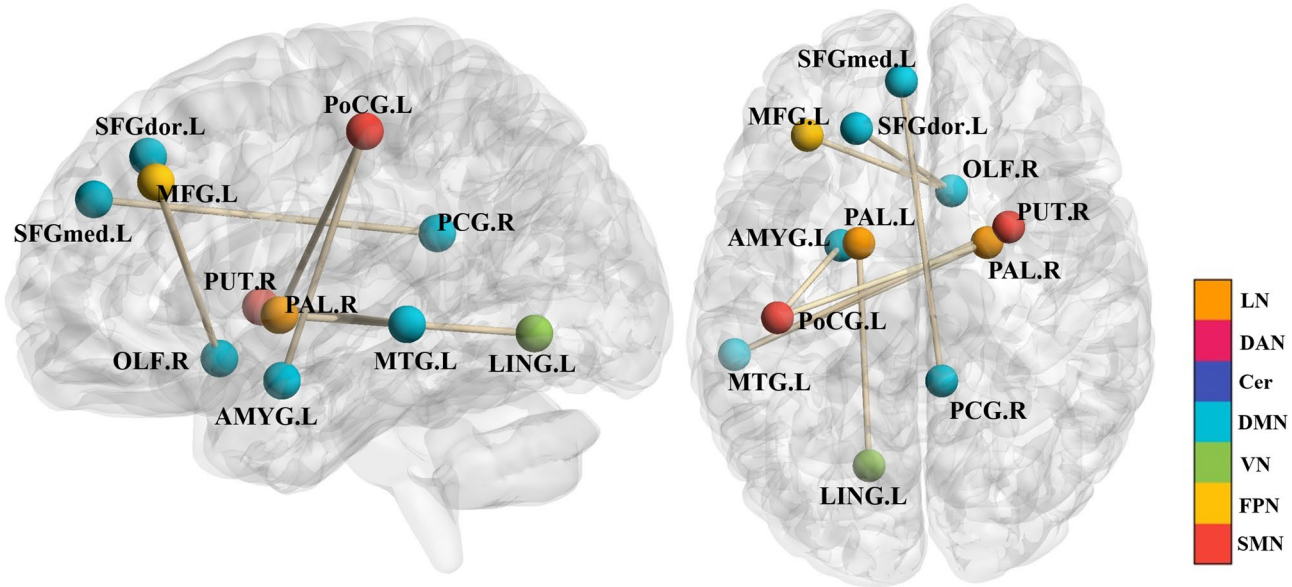


Fig. 8. The cerebral spatial distribution of the FC that contribute highly to the result prediction and the brain networks to which they belong.

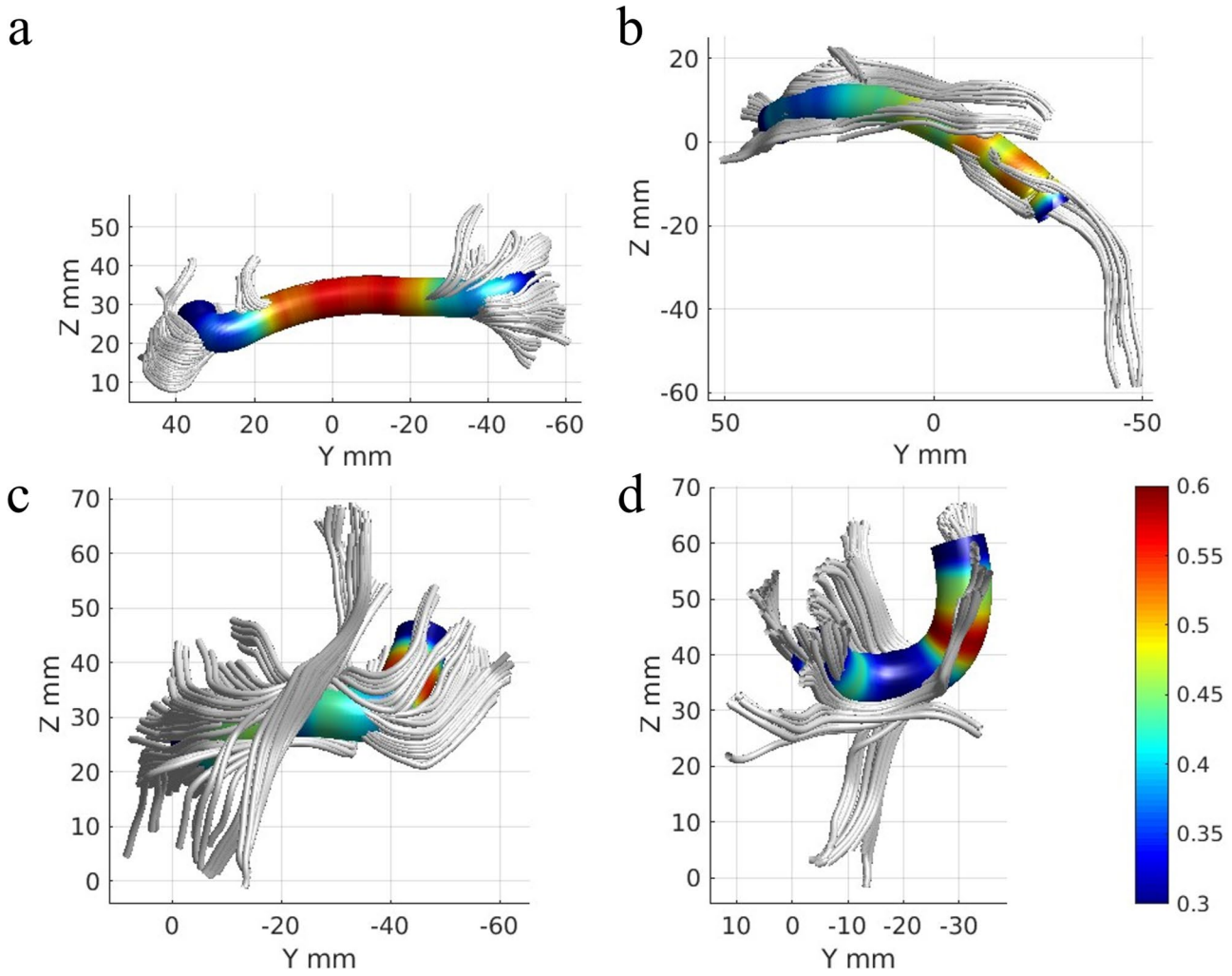


Fig. 9. The fiber tracts where the DTI features with high contributions to the result prediction are located, and the color band represents their FA values. (**a**) Left CGC; (**b**) Right TR; (**c**) Right SLF; (**d**) Left SLF.

lobe, occipital lobe and temporal lobe. The SLF is associated with functions such as spatial attention, vision and eye movement[53]. It can supply the prefrontal cortex with information from the parietal cortex regarding visual spatial perception. Meanwhile, the working memory of the prefrontal cortex also transmits information to the parietal cortex through the SLF, thus enabling the screening and retrieval of spatial information[54,55]. In addition, the structural integrity of the WM in the SLF is intimately associated with the cognitive ability necessary for the completion of memory tasks[56].

In addition to the traditional neuroimaging metrics extracted from the fMRI and DTI modalities, this study also adopted radiomics features based on the sMRI modality. These features aim to characterize the latent patterns in the images by extracting a large amount of high-throughput data from the neuroimages. sMRI can provide high-resolution images, showing in detail the anatomical structure of the brain. Extracting radiomics features from it can capture more subtle changes in the brain structure. After LASSO screening, the radiomics features obtained in this study belong to wavelet transform features and deep learning features. These features extract the multi-scale and multi-directional information in the images through wavelet transform[57]. The extraction of radiomic features enhances the stability of the model's input and improves its ability to generalize by leveraging the richness of multi-dimensional feature sources. In this way, the model can more effectively capture potential and non-traditional imaging information features, thus helping to identify the differential markers between the flight population and the general population, which may be difficult to detect in traditional diagnostic metrics.

By analyzing the high-contribution features obtained from the SHAP values and combining with the physiological meanings of the modalities corresponding to each feature, we found that these features with high contributions to the prediction results all originated from the brain regions or structures that are closely related to advanced cognitive functions, visual processing, and attention allocation. This may indicate that after flight training, flight cadets have differences in cognitive function, visual processing, and attention control compared with ordinary ground cadets. We speculate that this result may be closely related to long-term flight training. Flight training requires cadets to quickly integrate and screen various forms of information in a complex and

dynamic environment, which requires cadets to always maintain a high level of concentration and reasonable attention allocation.

In addition, for the machine learning models, we found that GNB and LR performed the best overall in classifying the dataset. GNB and LR are suitable for situations where there is a simple linear relationship between features and have poor classification performance for high-dimensional and complex data[58,59], while SVM and RF, which are suitable for solving non-linear relationships, high-dimensional data, and large-scale data, performed relatively poorly. Therefore, we speculate that the features screened out based on the functional MRI data of flight cadets may have a simple linear correlation, but the specific correlation pattern requires further research in combination with the neurophysiological mechanism.

There are still some shortcomings in this study. Our sample size is small, and to achieve better generalization ability, we need more participants. Furthermore, future studies may consider supplementing an independent comparison between sMRI morphological features and radiomic features to explore whether the utilization of radiomic features yields superior outcomes. Based on the SHAP analysis results of the present study, subsequent research can focus on models constructed exclusively with DTI and fMRI features, thereby enabling a more in-depth investigation into the alterations and impacts of flight training on brain structure and function.

Conclusion

This study successfully developed and validated four machine learning models based on flight cadets' sMRI, DTI, and fMRI data. Eventually, the sMRI + DTI + fMRI + LR model demonstrated the best prediction performance (accuracy: 0.838, AUC: 0.942, sensitivity: 0.835, specificity: 0.834). The model developed in this study can effectively distinguish between flight cadets and the general population. Among the developed machine learning models, the features with higher SHAP values may be associated with the biological markers related to flight ability, which provides potential clues for further scientific research on the brains of pilots. In addition, the effectiveness of this model offers a new perspective that combines machine learning and neuroimaging for the training evaluation and selection of flight cadets.

Data availability

The data and code of this study have been archived in FigShare. The access link to the dataset is: https://doi.org/10.6084/m9.figshare.28608278.v1.

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Author contributions

L.Y. designed the experiment, provided funding support, drafted the manuscript, and approved the final version. S.H.W. collected and analyzed the data, and contributed to the manuscript drafting. L.Y.B. contributed to the analysis and interpretation of the data. D.F.Y. provided funding support and supervised the experiment. H.Q.S. reviewed the manuscript. All authors participated in the study and approved the final version of the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

Ethics statement
The studies involving human participants were reviewed and approved by the Ethics Committee of the University of Electronic Science and Technology of China. The participants provided their written informed consent to take part in this study.

Additional information
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